Human Development Index (HDI) Predictor

1. INTRODUCTION

1.1 Project Title

Human Development Index (HDI) Predictor

1.2 Project Overview

The **HDI Predictor** is a web-based machine learning application that predicts the Human Development Index (HDI) score of countries based on key development indicators. The application uses Linear Regression to analyze four primary features:

1. **Life Expectancy** (years)
2. **Mean Years of Schooling**
3. **Expected Years of Schooling**
4. **GNI per Capita** (USD PPP)

The system provides:

- Instant HDI score predictions
- Category classification (Very High, High, Medium, Low)
- Interactive data visualizations
- Country-specific analysis
- REST API for integration

2. PROJECT OVERVIEW

2.1 Purpose

The primary purpose of the HDI Predictor is to democratize access to human development analytics by providing a simple, accessible tool for:

- **Policymakers**: Making data-driven decisions for resource allocation
- **Researchers**: Analyzing development patterns and trends
- **Development Organizations**: Identifying countries requiring intervention
- **General Public**: Understanding human development concepts

2.2 Features

Feature	Description
Interactive Prediction	Enter 4 indicators and get instant HDI prediction
Country Selection	Select from 30+ countries with preset values
Quick Test Values	Pre-set values for Very High, High, Medium, Low categories
Data Visualizations	Distribution plots, heatmaps, scatter plots
Category Classification	Auto-classification into 4 HDI tiers

3. IDEATION PHASE

3.1 Brainstorming & Idea Prioritization

3.1.1 Initial Ideas Generated

Idea ID	Idea Description	Category
IDEA-01	HDI Score Prediction using Machine Learning	Prediction
IDEA-02	Country-wise HDI Comparison Dashboard	Visualization
IDEA-03	Real-time HDI Monitoring System	Monitoring
IDEA-04	AI-powered Development Recommendations	Advisory
IDEA-05	Interactive Data Visualization Tool	Visualization

3.1.2 Idea Grouping

Group	Ideas Included	Theme
Group A	IDEA-01, IDEA-02, IDEA-07	Prediction & Analysis
Group B	IDEA-03, IDEA-04, IDEA-08	Monitoring & Advisory
Group C	IDEA-05, IDEA-06	Visualization & Mobile

3.2 Customer Problem Statement

3.2.1 For Government Policymakers

Element	Description
I am	A government policymaker responsible for national development planning
I'm trying to	Measure and track my country's human development progress
But	I lack accessible tools to predict HDI scores based on available indicators
Because	Traditional methods are slow, expensive, and require specialized expertise
Which makes me feel	Frustrated and uncertain about resource allocation and policy effectiveness

3.3 Empathy Map Canvas

User Persona: Government Policymaker in a Developing Country

Think & Feel

- Wants to make data-driven policy decisions
- Concerned about national development progress
- Feels pressure to allocate resources effectively
- Believes in sustainable development goals
- Wants to compare with neighboring countries

Hear

- Development reports from international organizations

- Feedback from citizens and local communities
- Expert opinions from economists and researchers
- Criticism about government performance

See

- Economic growth indicators
- Education and healthcare statistics
- International development rankings
- Social media discussions about development
- Budget allocation documents

Say & Do

- Advocates for data-driven policy making
- Attends international development summits
- Allocates budget for development programs
- Monitors progress through reports
- Collaborates with international organizations

Pain

- Limited access to predictive analytics tools
- Uncertainty about policy effectiveness
- Resource constraints and budget limitations
- Pressure from political stakeholders
- Difficulty in measuring intangible development aspects

Gain

- Better resource allocation decisions
- Improved development outcomes
- Enhanced international standing
- Informed policy making
- Increased citizen satisfaction

- # 4. PROJECT DESIGN PHASE
- ## 4.1 Problem-Solution Fit Template

Aspect	Description
Customer Segment	Government policymakers, researchers, development organizations
Customer Problem	Lack of accessible, accurate tools for predicting HDI scores
Alternative Solutions	1. Manual statistical analysis 2. International organization reports (UNDP) 3. Static datasets without predictive capabilities
Value Proposition	Accessible, web-based HDI prediction tool using Machine Learning for accurate time development assessment
Solution Features	1. ML-powered HDI score prediction 2. Interactive data visualizations 3. Country comparison capabilities 4. User-friendly interface 5. Comprehensive reports
Success Metrics	1. Prediction accuracy (R^2 > 0.85) 2. User satisfaction rate 3. Number of predictions made 4. Time saved in analysis

- ## 4.2 Proposed Solution Template

S.No.	Parameter	Description
1.	**Problem Statement**	Decision-makers lack an accessible, data-driven tool to pre Human Development Index scores based on key indicators

S.No.	Parameter	Description
2.	**Idea/Solution Description**	Web-based application powered by Machine Learning (Lin Regression) predicting HDI scores based on 4 key indicato
3.	**Novelty/Uniqueness**	• Combines Machine Learning with HDI analytics • Interactive visualizations with multiple chart types • Real-time prediction capabilities • Country selection with preset values • REST API for integration
4.	**Social Impact/Customer Satisfaction**	• Enables data-driven policy decisions • Supports sustainable development goals • Helps identify countries needing intervention • Promotes transparency in development assessment
		• Freemium model: Basic free, premium paid

5.2 Component Description

Component	**Technology**	**Purpose**
User Interface	HTML5, CSS3, Bootstrap 5	Responsive frontend
Web Framework	Flask (Python)	Backend routing and logic
ML Model	Scikit-learn (Linear Regression)	Prediction engine
Data Processing	Pandas, NumPy	Data manipulation
Data Visualization	Matplotlib, Seaborn	Chart generation
Serialization	Pickle	Model storage and loading

Component	Technology	Purpose
IDE	Visual Studio Code	Development environment

6. SOLUTION REQUIREMENTS

6.1 Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	**User Input**	User can enter Life Expectancy value
FR-2	**User Input**	User can enter Mean Years of Schooling value
FR-3	**User Input**	User can enter Expected Years of Schooling value
FR-4	**User Input**	User can enter GNI per Capita value
FR-5	**Country Selection**	User can select a country from dropdown
FR-6	**Prediction**	System predicts HDI score using ML model
FR-7	**Categorization**	System categorizes HDI as Very High, High, Medium, or Low
FR-8	**Visualization**	System displays data visualizations (distribution, heatmap, sca plots)

7.1 Data Flow Diagram (DFD)

Level 0: Context Diagram

```
┌────────────────┐
|        |
|  USER INPUT   |
```

```
| (Life Expectancy)|
| (Schooling)     |
| (GNI per Capita)|
└─────────┬─────────┘
┌──────────────────┐
|              |
|  HDI PREDICTOR
└─────────┬─────────┘
┌──────────────────┐
|              |
|   OUTPUT      |
| (HDI Score)     |
| (Category)     |
| (Visualizations)|
└──────────────────┘
```

8. PROJECT PLANNING

Project Timeline (Gantt Chart)

```
Task                    | Week 1 | Week 2 | Week 3 | Week 4 | Week 5 |
----------------------------|--------|--------|--------|--------|--------|
1. Project Setup          |████████|        |        |        |        |
2. Data Preparation       |        |████████|        |        |        |
3. Model Development      |        |        |████████|        |        |
4. Flask Application      |        |        |        |████████|        |
5. Integration & Testing  |        |        |        |        |████████|
```

9. Testing Environment

Parameter	Value
URL/Location	http://127.0.0.1:5000
Browser	Chrome, Firefox, Edge, Safari
Credentials	None Required

Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-001	Home Page Load	1. Open browser 2. Navigate to URL 3. Verify home page loads	Home page displays correctly with introduction	✅ Pass
TC-002	Navigation	1. Click "Predict" button 2. Click "Visualizations" link 3. Click "About" link	All pages navigate correctly	✅ Pass
TC-003	Prediction Form - Valid Input	1. Enter life expectancy: 80.0 2. Enter mean schooling: 14.0 3. Enter expected schooling: 16.0 4. Enter GNI: 50000 5. Click "Predict"	HDI score ~0.900+ predicted	✅ Pass

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-004	Prediction Form - Invalid Input	1. Enter negative values 2. Click "Predict"	Error message displayed	✓ Pass
TC-005	Country Selection	1. Select "Norway" from dropdown 2. Click "Predict"	Prediction uses country name	✓ Pass
TC-006	Quick Test - Very High	1. Click "Very High" button 2. Click "Predict"	HDI > 0.800 predicted	✓ Pass
TC-007	Quick Test - High	1. Click "High" button 2. Click "Predict"	HDI 0.700-0.799 predicted	✓ Pass

10. REFERENCES

10.1 Online Resources

Resource	URL	Description
Mural Templates	https://www.mural.co/templates/brainstorm-and-idea-prioritization	Brainstorming templates
Miro Templates	https://miro.com/templates/customer-problem-statement/	Problem statement templates
Flask Documentation	https://flask.palletsprojects.com/	Flask web framework
Scikit-learn	https://scikit-learn.org/	Machine learning library

Resource	URL	Description
Pandas	https://pandas.pydata.org/	Data manipulation
Matplotlib	https://matplotlib.org/	Visualization library
Seaborn	https://seaborn.pydata.org/	Statistical visualization
UNDP HDI Data	https://hdr.undp.org/data-center/human-development-index	Official HDI data
Kaggle Dataset	https://www.kaggle.com/datasets/undp/human-development-index	HDI dataset

CONCLUSION

Project Summary

The **Human Development Index (HDI) Predictor** is a complete machine learning web application that successfully demonstrates:

Aspect	Achievement
Problem Solved	Provides accessible HDI prediction tool for policymakers, researchers, and development organizations
Technical Implementation	Linear Regression model with R^2 Score of ~0.90 (90% accuracy)
Web Interface	User-friendly Flask application with 4 interactive pages
Visualizations	5 different data visualizations for comprehensive analysis

https://github.com/datlacharan1234-rcb/hdi_predicter.git